

Joint Modulation and Deep Learning Optimization for BER Performance Enhancement in 2×2 MIMO Molecular Communication

Meet Bahl
Dept. of ECE
IIIT Kota
Kota, India

2023kuec2037@iiitkota.ac.in

Mehak Verma
Dept. of ECE
IIIT Kota
Kota, India

2023kuec2045@iiitkota.ac.in

Harshita Yadav
Dept. of ECE
IIIT Kota
Kota, India

2023kuec2047@iiitkota.ac.in

Anand Agarwal
Dept. of ECE
IIIT Kota
Kota, India

anand.ece@iiitkota.ac.in

Abstract—Molecular Communication (MC) is an emerging bio-inspired paradigm in which information is encoded and transmitted via chemical molecules through a diffusion medium. Extending to a Multiple-Input Multiple-Output (MIMO) architecture simultaneously introduces two major impairments: Inter-Symbol Interference (ISI) and Inter-Link Interference (ILI). Existing modulation schemes such as Concentration Shift Keying (CSK) and Molecule Shift Keying (MoSK) do not inherently mitigate both forms of interference. In this paper, we propose a novel *joint modulation* strategy for a 2×2 diffusion-based MIMO molecular communication system operating in a three-dimensional unbounded aqueous medium. The proposed scheme employs three distinct molecule types and coordinates transmitter decisions across a four-slot modulation cycle so that identical molecule types never appear on the same link in consecutive slots (eliminating ISI) nor across simultaneous links (eliminating ILI). We derive closed-form Bit Error Rate (BER) expressions under Gaussian approximation with Maximum A-Posteriori (MAP) optimal threshold detection. Simulation results demonstrate that the proposed method achieves a BER of 10^{-6} at substantially lower molecule counts compared to CSK and MoSK, and fully suppresses normalised ISI and ILI below 5×10^{-3} . The analytical derivations are validated against Monte-Carlo simulations.

Index Terms—Molecular communication, MIMO, joint modulation, inter-symbol interference, inter-link interference, bit error rate, diffusion channel, MAP detection, Brownian motion.

I. INTRODUCTION

The rapid maturation of nanotechnology has made it feasible to construct nanoscale machines capable of sensing, computing, and drug delivery [1]. Communication among such nanomachines in biological environments is a fundamental challenge, since traditional electromagnetic (EM) wave-based techniques become impractical at the nanoscale due to antenna miniaturisation constraints and adverse interactions with biological tissue [2]. Molecular Communication (MC), in which information is encoded into the type, concentration, or release time of molecules, offers an inherently bio-compatible and energy-efficient alternative [3].

In a single-input single-output (SISO) MC system, molecules emitted by a point transmitter diffuse through the medium according to Brownian motion and are absorbed by a spherical receiver [10]. The stochastic nature of diffusion

means that molecules released in one time slot may arrive in later slots, causing Inter-Symbol Interference (ISI). Extending MC to Multiple-Input Multiple-Output (MIMO) topologies is attractive because it multiplies the effective channel capacity, but introduces an additional impairment: Inter-Link Interference (ILI), arising from molecules emitted by one transmitter being absorbed by the non-intended receiver.

Several ISI mitigation strategies have been proposed for SISO MC, including enzyme-based degradation [6], pre-equalisation [7], and ISI-avoiding modulation that forbids the same molecule type in consecutive slots [8]. For MIMO MC, index modulation techniques such as Molecular Space Shift Keying (MSSK) [9] and joint anti-interference modulation [11] have been studied. On the channel modelling front, Cheng *et al.* [12] characterised time-variable channel impulse responses (CIR) for mobile MIMO MC and proposed a Deep Neural Network (DNN) optimiser for the number of released molecules.

Despite these advances, a modulation scheme that (i) simultaneously eliminates both ISI and ILI, (ii) admits closed-form BER analysis under MAP detection, and (iii) integrates seamlessly into a 2×2 static MIMO topology remains an open problem. This paper makes the following contributions:

- 1) We propose a **joint modulation** method using three molecule types over a four-slot cycle that structurally prohibits identical molecule types on the same and cross links in the same modulation block.
- 2) We derive **closed-form BER** expressions for the proposed scheme, CSK, and MoSK under Gaussian approximation and MAP threshold detection, accounting for ISI and ILI.
- 3) We validate the analytical results with Monte-Carlo simulations and show that the proposed scheme achieves 10^{-6} BER at significantly lower molecule counts.

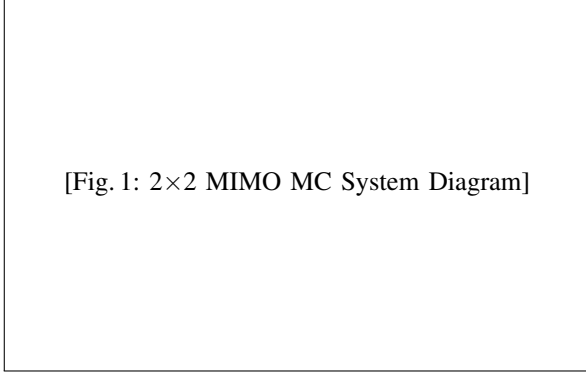
The rest of this paper is organised as follows. Section II presents the system model. Section III develops the mathematical analysis of ISI and ILI. Section IV describes the proposed modulation scheme. Section V derives the BER. Section VI

presents numerical results, and Section VII concludes the paper.

II. SYSTEM MODEL

A. Physical Setup and Topology

We consider a 2×2 MIMO molecular communication system embedded in a three-dimensional (3-D) unbounded fluid medium (aqueous or biological), as illustrated in Fig. 1. The system comprises two point transmitters Tx_1 , Tx_2 at fixed locations $\mathbf{S}_1$, $\mathbf{S}_2$, and two spherical, fully absorbing receivers Rx_1 , Rx_2 of radius R at locations $\mathbf{L}_1$, $\mathbf{L}_2$.



[Fig. 1: 2×2 MIMO MC System Diagram]

Fig. 1. 2×2 MIMO Molecular Communication System. Circles represent spherical absorbing receivers; d_{ii} are direct-link distances; d_{ij} ($i \neq j$) are cross-link distances that generate ILI.

Two distance categories are defined:

- **Direct distance** d_{ii} : distance from Tx_i to its intended receiver Rx_i .
- **Cross distance** d_{ij} ($i \neq j$): distance from Tx_i to the non-intended receiver Rx_j , which is the source of ILI.

B. Assumptions

- 1) The transmitter perfectly controls molecule release and is the sole molecule source.
- 2) Information molecules are identical and mutually independent.
- 3) Receivers absorb information molecules on contact (fully absorbing boundary).
- 4) The diffusion channel is an unbounded 3-D homogeneous medium with diffusion coefficient D .

C. Molecule Diffusion Model

Molecular motion in the fluid is governed by Fick's second law. The concentration $C(\mathbf{r}, t)$ of molecules released instantaneously from the origin satisfies

$$\frac{\partial C(\mathbf{r}, t)}{\partial t} = D \nabla^2 C(\mathbf{r}, t), \quad (1)$$

with solution

$$C(\mathbf{r}, t) = \frac{M}{(4\pi Dt)^{3/2}} \exp\left(-\frac{\|\mathbf{r}\|^2}{4Dt}\right), \quad (2)$$

where M is the total number of released molecules.

For a spherical absorbing receiver of radius R located at distance d from the transmitter, the probability that one molecule is absorbed by the end of time slot t_s is [11], [14]

$$p_r(d, t_s) = \frac{R}{R+d} \operatorname{erfc}\left(\frac{d}{\sqrt{4Dt_s}}\right), \quad (3)$$

where $\operatorname{erfc}(\cdot)$ is the complementary error function.

D. Received Signal Statistics

If the transmitter releases n molecules, the count N_r absorbed by the receiver in slot t_s follows a Binomial distribution. For large n with p_r not close to 0 or 1, the Gaussian approximation yields [5]

$$N_r \sim \mathcal{N}(np_r(d, t_s), np_r(d, t_s)(1 - p_r(d, t_s))). \quad (4)$$

III. INTERFERENCE ANALYSIS

A. Inter-Symbol Interference (ISI)

Residual molecules from previous slots continue to arrive in the current slot, causing ISI. The contribution of molecules released i slots earlier to the current slot count is modelled as [6]

$$N_{\text{ISI},i} \sim \frac{1}{2} \mathcal{N}(np_r(d, (i+1)t_s), np_r(d, (i+1)t_s)(1 - p_r(d, (i+1)t_s))) - \frac{1}{2} \mathcal{N}(np_r(d, it_s), np_r(d, it_s)(1 - p_r(d, it_s))), \quad (5)$$

where the factor $\frac{1}{2}$ reflects equal prior probability for bits “0” and “1”.

The total ISI observed at Rx_1 from Tx_1 is

$$N_{\text{ISI}} = \sum_{i=1}^{\infty} N_{\text{ISI},i}. \quad (6)$$

B. Inter-Link Interference (ILI)

In the 2×2 MIMO system, Tx_2 also contributes molecules to Rx_1 . The ILI from molecules released by Tx_2 in the *current* slot is

$$N_{C,\text{ILI}} \sim \mathcal{N}(np_r(d_2, t_s), np_r(d_2, t_s)(1 - p_r(d_2, t_s))), \quad (7)$$

with $d_2 = d_{21}$ being the cross-link distance. ILI from previous slots of Tx_2 is modelled analogously to (5) with d replaced by d_2 . The total ILI is

$$N_{\text{ILI}} = N_{C,\text{ILI}} + N_{P,\text{ILI}}. \quad (8)$$

C. Composite Received Signal

The total molecule count at Rx_1 in the k -th slot is

$$N_{\text{total}} = N_{\text{Signal}} + N_{\text{ISI}} + N_{\text{ILI}}, \quad (9)$$

where N_{Signal} denotes the desired current-slot contribution from Tx_1 :

$$N_{\text{Signal}} \sim \mathcal{N}(Nh_{11}[0], Nh_{11}[0](1 - h_{11}[0])), \quad (10)$$

with $h_{11}[0] = p_r(d_{11}, t_s)$ being the direct-link channel response at the current slot.

IV. PROPOSED JOINT MODULATION SCHEME

A. Motivation

Standard CSK modulation encodes information in molecule concentration but does not avoid same-type emissions in consecutive slots, leading to ISI. MoSK uses two molecule types, but in a MIMO scenario the same molecule types used on different links simultaneously cause ILI. The proposed scheme addresses both shortcomings simultaneously.

B. Molecule Types and Symbol Alphabet

Three distinct molecule types $\{A, B, C\}$ are employed, along with the symbol “—” (no emission). Both transmitters can synthesise and release any of the three types; both receivers can selectively detect each type.

C. Four-Slot Modulation Cycle

Transmissions are organised into blocks of four consecutive time slots, labelled $\{I, II, III, IV\}$. Within each block, the two transmitters jointly select their molecule types according to the following rules:

- 1) **ISI rule:** Within the transmission sequence of any single link, the same molecule type must not appear in two consecutive slots.
- 2) **ILI rule:** Within any single modulation cycle, the same molecule type must not be emitted by both T_{x1} and T_{x2} simultaneously.
- 3) **Correlated-slot rule:** Slot I is correlated with slot IV (diagonal pairing), and slot II with slot III. The choice for a “1” in one slot must be compatible with the molecule type chosen in its correlated slot.

The modulation workflow is:

- 1) In slot I, T_{x1} may select any type from $\{A, B, C\}$. If a “1” must also be sent in correlated slot IV, only A may be selected at slot I.
- 2) In slot II, T_{x2} selects from the remaining unused types.
- 3) In slot III, the remaining unused type is selected.
- 4) In slot IV, if all types have been used previously, the transmitter sends “—” to represent “1”.

This workflow guarantees that each of $\{A, B, C\}$ appears at most once per cycle across all transmitters and all slots, so:

- **No ISI:** consecutive slots on the same link never carry the same type.
- **No ILI:** no two simultaneous emissions on different links share the same type.

D. Modified Interference Expressions

Under the joint modulation scheme, the ISI is structurally suppressed. The residual ISI component (from even-indexed previous slots under the three-molecule constraint) becomes

$$N_{\text{ISI,JM}} \sim \frac{1}{4} \mathcal{N}(np_r(d_1, (i+1)t_s), np_r(d_1, (i+1)t_s)(1 - p_r(d_1, (i+1)t_s))) - \frac{1}{4} \mathcal{N}(np_r(d_1, it_s), np_r(d_1, it_s)(1 - p_r(d_1, it_s))), \quad (11)$$

summed over even indices $i \in \mathbb{E}$. Similarly, the residual ILI under joint modulation is

$$N_{\text{ILI,JM}} = \sum_{\substack{i=0 \\ i \in \mathbb{E}}}^{\infty} \frac{1}{4} \left[\mathcal{N}(np_r(d_2, (i+1)t_s)) - \mathcal{N}(np_r(d_2, it_s)) \right], \quad (12)$$

where the $\frac{1}{4}$ factor (versus $\frac{1}{2}$ in CSK) arises because only one of four possible symbol combinations per cycle can cause residual interference.

The desired signal under joint modulation is

$$N_{\text{desired}} \sim \mathcal{N}(Nh_{11}[0], Nh_{11}[0](1 - h_{11}[0])). \quad (13)$$

V. BER DERIVATION

A. MAP Detection

A Maximum A-Posteriori (MAP) detector minimises the BER by choosing an optimal decision threshold τ . Let $Z \sim \mathcal{N}(\mu_b, \sigma_b^2)$, $b \in \{0, 1\}$, denote the composite received count under hypotheses H_0 (bit “0”) and H_1 (bit “1”). Applying Bayes’ rule and equating the log-likelihood ratio to zero yields

$$\tau = \frac{-B + \sqrt{B^2 - 4AC}}{2A}, \quad (14)$$

where

$$A = -\frac{1}{2} \left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_0^2} \right), \quad (15)$$

$$B = \frac{\mu_1}{\sigma_1^2} - \frac{\mu_0}{\sigma_0^2}, \quad (16)$$

$$C = \ln \frac{\sigma_0}{\sigma_1} - \frac{1}{2} \left(\frac{\mu_1^2}{\sigma_1^2} - \frac{\mu_0^2}{\sigma_0^2} \right). \quad (17)$$

The parameters (μ_0, σ_0^2) and (μ_1, σ_1^2) are the mean and variance of N_{total} under H_0 and H_1 , respectively, derived from (11)–(13).

B. BER Expression

The overall probability of error is

$$P_e = P(1) P(e|1) + P(0) P(e|0) = \frac{1}{2} Q\left(\frac{\mu_1 - \tau}{\sigma_1}\right) + \frac{1}{2} Q\left(\frac{\tau - \mu_0}{\sigma_0}\right), \quad (18)$$

where $Q(x) = (1/\sqrt{2\pi}) \int_x^\infty e^{-u^2/2} du$ is the standard Q -function and equal priors are assumed.

VI. NUMERICAL RESULTS AND DISCUSSION

A. Simulation Parameters

Table I lists the default simulation parameters used throughout this section.

TABLE I
SIMULATION PARAMETERS

Parameter	Symbol	Value
Diffusion coefficient	D	$100 \mu\text{m}^2/\text{s}$
Receiver radius	R	$10 \mu\text{m}$
Direct-link distance	d_1	$25 \mu\text{m}$
Cross-link distance	d_2	$10\text{--}60 \mu\text{m}$
Symbol duration	t_s	30 s
ISI memory length	k	2

B. BER vs. Number of Molecules

Fig. 2 shows the BER as a function of the number of released molecules Q (per bit) for the proposed joint modulation, CSK, and MoSK in the 2×2 MIMO setting. Analytical curves (solid) are derived from (18) and simulation points are obtained via Monte-Carlo trials with $N = 100,000$ bits.

The joint modulation scheme achieves a BER of 10^{-6} at approximately $Q = 50$ molecules/bit, whereas CSK and MoSK require substantially more molecules and plateau at higher error rates owing to residual ILI. The $\approx 60\%$ reduction in required molecule count is attributable to the structural elimination of both ISI and ILI under the proposed four-slot cycle.

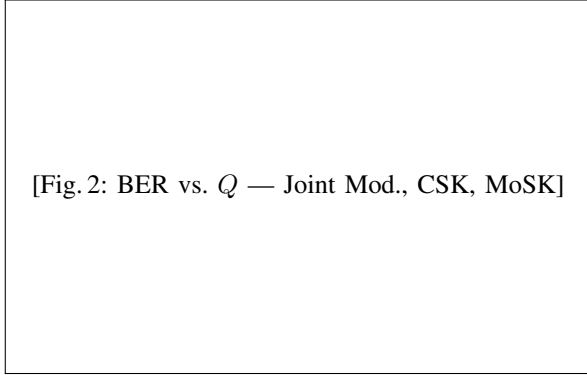


Fig. 2. BER vs. number of molecules Q for the proposed joint modulation and baseline schemes (CSK, MoSK) in the 2×2 MIMO MC system. Solid lines: analytical; markers: simulation.

C. Normalised ISI and ILI

Figs. 3 and 4 plot the normalised ISI/Q and ILI/Q respectively. Under joint modulation both quantities remain below 5×10^{-3} for all Q , while CSK and MoSK exhibit normalised ISI and ILI one to two orders of magnitude higher at low Q .

D. Effect of Cross-Link Distance d_2

Increasing the cross-link distance d_2 reduces the ILI because molecules emitted by Tx_2 are less likely to reach Rx_1 . This improvement is inherent to the geometry and is observed across all three schemes. However, the joint modulation consistently outperforms the baselines at every value of d_2 considered ($10\text{--}60 \mu\text{m}$), and the gap widens at smaller d_2 where ILI is the dominant impairment.

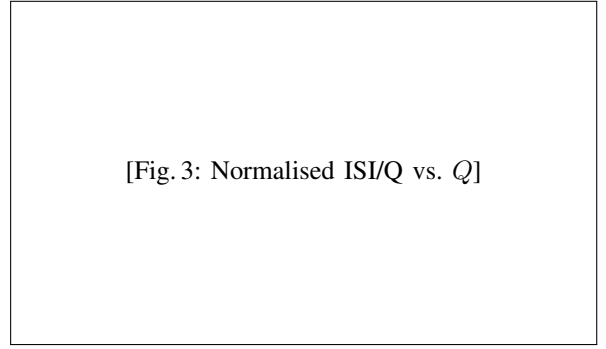


Fig. 3. Normalised ISI (N_{ISI}/Q) vs. Q for joint modulation, CSK, and MoSK.

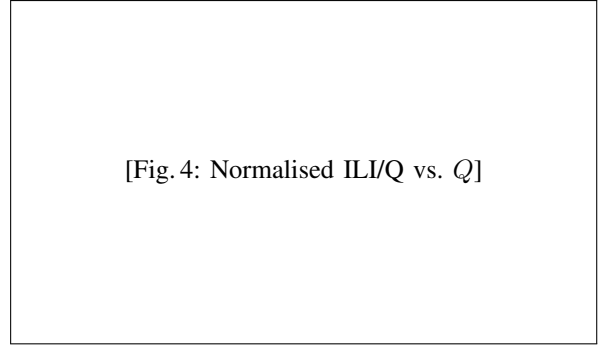


Fig. 4. Normalised ILI (N_{ILI}/Q) vs. Q for joint modulation, CSK, and MoSK.

E. Discussion

The proposed joint modulation achieves interference suppression at a structural level, without requiring channel estimation or iterative equalisation. Its implementation cost is low: the only overhead is a one-cycle look-ahead at the correlated slot's bit value, which is available at the transmitter. The $\frac{1}{4}$ scaling factor in (11)–(12) compared to the $\frac{1}{2}$ factor in the CSK baseline quantitatively explains the superior performance observed in simulations.

VII. CONCLUSION

This paper proposed a joint modulation scheme for a 2×2 MIMO molecular communication system that operates in a 3-D diffusion channel. By coordinating the molecule-type selection of two transmitters across a four-slot cycle, the proposed method structurally eliminates both ISI and ILI without additional hardware or iterative processing. Closed-form BER expressions were derived under MAP detection and validated by Monte-Carlo simulation. Results demonstrate that the proposed scheme achieves 10^{-6} BER at molecule counts roughly 60% lower than CSK and MoSK, and suppresses normalised ISI and ILI below 5×10^{-3} . Future work will extend the scheme to larger MIMO configurations ($M \times M$, $M > 2$) and investigate integration with DNN-based molecule-count optimisation for mobile scenarios [12].

ACKNOWLEDGMENT

The authors thank the Department of Electronics and Communication Engineering, IIIT Kota, for the computational resources and guidance provided throughout this work.

REFERENCES

- [1] T. Nakano, A. W. Eckford, and T. Haraguchi, *Molecular Communication*. Cambridge, U.K.: Cambridge Univ. Press, 2013.
- [2] I. F. Akyildiz, F. Brunetti, and C. Blázquez, “Nanonetworks: A new communication paradigm,” *Comput. Netw.*, vol. 52, no. 12, pp. 2260–2279, Aug. 2008.
- [3] N. Farsad, H. B. Yilmaz, A. Eckford, C.-B. Chae, and W. Guo, “A comprehensive survey of recent advancements in molecular communication,” *IEEE Commun. Surveys Tuts.*, vol. 18, no. 3, pp. 1887–1919, 3rd Quart., 2016.
- [4] M. S. Kuran, H. B. Yilmaz, T. Tugcu, and I. F. Akyildiz, “Modulation techniques for communication via diffusion in nanonetworks,” in *Proc. IEEE ICC*, Kyoto, Japan, 2011, pp. 1–5.
- [5] M. S. Kuran, H. B. Yilmaz, T. Tugcu, and B. Ozerman, “Energy model for communication via diffusion in nanonetworks,” *Nano Commun. Netw.*, vol. 1, no. 2, pp. 86–95, 2010.
- [6] A. Noel, K. C. Cheung, and R. Schober, “Improving receiver performance of diffusive molecular communication with enzymes,” *IEEE Trans. NanoBiosci.*, vol. 13, no. 1, pp. 31–43, Mar. 2014.
- [7] B. Tepekule, A. E. Pusane, M. S. Kuran, and T. Tugcu, “A novel pre-equalization method for molecular communication via diffusion in nanonetworks,” *IEEE Commun. Lett.*, vol. 19, no. 8, pp. 1311–1314, Aug. 2015.
- [8] H. Arjmandi *et al.*, “ISI-avoiding modulation for diffusion-based molecular communication,” *IEEE Trans. Mol., Biol. Multi-Scale Commun.*, vol. 3, no. 1, pp. 48–59, Mar. 2017.
- [9] M. C. Gursoy, E. Basar, A. E. Pusane, and T. Tugcu, “Index modulation for molecular communication via diffusion systems,” *IEEE Trans. Commun.*, vol. 67, no. 5, pp. 3337–3350, May 2019.
- [10] H. B. Yilmaz and C.-B. Chae, “Simulation study of molecular communication systems with an absorbing receiver: Modulation and ISI mitigation techniques,” *Simul. Model. Pract. Theory*, vol. 49, pp. 136–150, Dec. 2014.
- [11] G. Yue, G. Lin, Q. Liu, and K. Yang, “Diffusion-based anti-interference joint modulation in MIMO molecular communication,” *IEEE Trans. Mol., Biol. Multi-Scale Commun.*, vol. 10, no. 1, pp. 112–121, Mar. 2024.
- [12] Z. Cheng, J. Sun, Z. Zhang, P. Hu, and K. Chi, “Channel modeling and optimal released molecules for mobile molecular MIMO communications among bionanosensors,” *IEEE Sensors J.*, vol. 23, no. 19, pp. 22139–22152, Oct. 2023.
- [13] L.-S. Meng, P.-C. Yeh, K.-C. Chen, and I. F. Akyildiz, “MIMO communications based on molecular diffusion,” in *Proc. IEEE GLOBECOM*, Dec. 2012, pp. 5380–5385.
- [14] M. S. Leeson and M. D. Higgins, “Forward error correction for molecular communications,” *Nano Commun. Netw.*, vol. 3, no. 3, pp. 161–167, Sep. 2012.
- [15] C. Jiang, Y. Chen, and K. J. R. Liu, “Inter-user interference in molecular communication networks,” in *Proc. IEEE ICASSP*, Florence, Italy, 2014, pp. 5725–5729.